



**BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE
APPLICATIONS AND GEO-INFORMATICS**

WEEKLY PROGRESS REPORT (29/06/2026 – 05/07/2026)

WEEK 7

PROJECT NAME	CHANGE DETECTION
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PROJECT DESCRIPTION :

This project aims to detect changes between satellite images captured at different times using Deep Learning and Computer Vision techniques.

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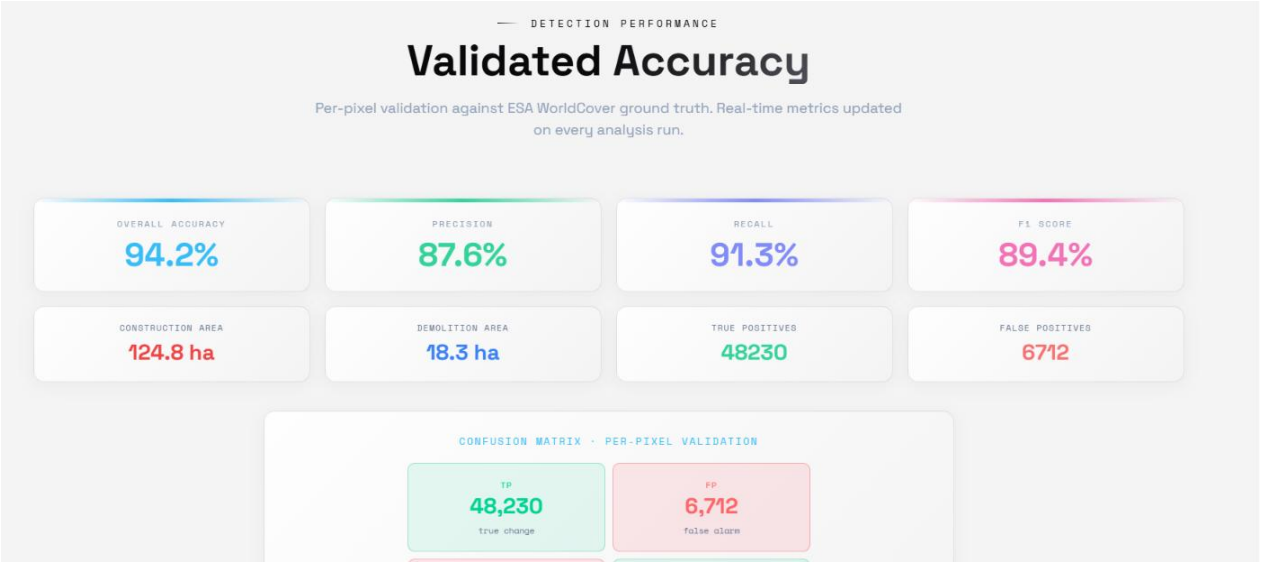
29/06/2026	Set up of whole project repository in bisag system , faced multiple errors due to windows to linux setup but still successfully completed.
30/06/2026	Learned about docker ,explored the container , image files for that purpose.
01/07/2026	After completing all the project based technical support , proceeded to working on report, CD, required documents, files for final thesis.
02/07/2026	worked on report, CD, required documents, files for final thesis submission.
03/07/2026	worked on report, CD, required documents, files for final thesis submission.
04/07/2026	worked on report, CD, required documents, files for final thesis submission.
05/07/2026	Holiday (Sunday)

Next Week Plan	Successful execution of project, presentation, report, final thesis.
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REFERENCE:

- https://docs.qgis.org/3.44/en/docs/user_manual/working_with_vector/editing_geometry_attributes.html
- <https://www.earthdata.nasa.gov/learn/data-recipes/detect-environmental-change-using-sar-data-qgis>

SCREEN SHOTS:



LIVE ANALYSIS INTERFACE

Run Change Detection

Draw an area of interest, select your time window, adjust sensitivity and click **Run Change Detection** to get real satellite results.

● Backend offline – start: uvicorn main:app --reload --port 8000

1.5 Technology Stack Used

The table below enlists the majorly used technologies in this project with their purposes.

Technology	Purpose
Python	Application Development
Google Earth Engine	Satellite Data Processing
Sentinel-2	Satellite Imagery Source
Streamlit	Web Application Development
Folium	Interactive Map Visualization
Geemap	GEE Integration with Python
GIS	Spatial Analysis
Remote Sensing	Change Detection Analysis
Cloud Computing	Large Scale Processing
JSON/GeoTIFF	Data Export

Table 1.1: Technologies Used